Kacem Mathlouthi

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Education

National Institute of Applied Science and Technology (INSAT) *Sep 2022 – Dec 2027*

Engineering Degree in Software Engineering

Relevant Coursework: Data Structures & Algorithms, Algorithmic Complexity, Object-Oriented Programming, Operating Systems, Computer Networks, Distributed Systems, SQL & NoSQL Databases, Software Architecture, Software Testing, Compilers, Machine Learning, Deep Learning, Big Data, MLOps

Experience

Software Engineer | *Callab AI (YC P26)* *Tunis, Tunisia (On Site)*

TypeScript, Python, React, Bun, AI SDK, MCP, PostgreSQL, Redis, Docker, Kubernetes, AWS *Jun 2025 – Present*

- **Shipped** features **end to end** across backend, AI, frontend and DevOps, and **owned production systems** in a multi-service platform, delivering dozens of features and resolving production incidents and bugs.
- **Built** the **Omnichannel AI Agent** serving Slack, Telegram, Teams, WhatsApp and web, with **human-in-the-loop approvals**, a flow engine, **long-term memory**, scheduled tasks, knowledge-base retrieval, MCP tools and skills, and conversation compaction.
- **Delivered human call supervision** end to end, letting operators watch live calls, listen in and **take over** from the **AI agent** mid-conversation, with agents escalating to a human when they cannot resolve a call.
- **Built** an **in-product AI assistant** that answers product questions and drives the app for the user, **reading and acting** on the **live DOM** through a browser tool family, so it can navigate, fill and complete tasks on the user's behalf.

Google Summer of Code Developer | *NRNB (National Resource for Network Biology)* *Remote*

Python, TypeScript, FastAPI, Next.js, Qdrant, Langfuse *May 2025 – Sep 2025*

- Collaborated with the VCell team to **develop and deploy VCell-AI**, an **AI agent platform** with tool calling capabilities, built on a Next.js frontend and FastAPI backend, expected to let **∼1,000 researchers** query, explore, and create biological models.
- **Integrated** a **RAG** system using **Qdrant** to provide contextual information about VCell software, tutorials, and guidelines, with **LLM observability** via **Langfuse** to track agent interactions, monitor performance, and improve reliability.
- **Enabled** the AI agent to **analyze diagrams and VCML files** using **multimodal** capabilities, supporting bio-model interpretation.

Machine Learning Engineering Intern | *Orange* *Tunis, Tunisia (On Site)*

Python, PyTorch, CrewAI, Streamlit, Flask *Aug 2024 – Oct 2024*

- **Developed** a lightweight **CNN model** for plant disease classification optimized for **edge devices** using **knowledge distillation**.
- **Built AI agentic workflow** for **automated report generation**, streamlining data collection, analysis, and research extraction.
- **Achieved 4th place** in **Orange Summer Challenge**, pitching the solution to both technical and non-technical stakeholders.

Publications

Generalization Measures under Controlled Covariate Shift: A Regime-Aware Benchmark *OpenReview*

Accepted at Transactions on Machine Learning Research (TMLR), 2026

- Evaluated 40+ generalization measures on CIFAR-10-C/P under controlled covariate shift, finding that predictivity is strongly regime-dependent and that IID performance does not reliably transfer to distribution shift or model selection.

Projects

Animus: *TypeScript, React, Hono, Bun, AI SDK, Daytona, Manim, PostgreSQL, R2* *Live* | *GitHub*

- **Built and deployed** an interactive **AI agent** that researches topics and iteratively **writes, renders, diagnoses, and repairs** narrated **explainer videos** in persistent **Daytona sandboxes**.
- **Engineered** a **tool-calling workflow** with **ElevenLabs narration**, FFmpeg music mixing, R2 storage, and public share links.

Metis AI: *FastAPI, React, Celery, Redis, PostgreSQL, GitHub App, GCP* *GitHub*

- **Built and deployed** a full-stack **GitHub App** that **automates pull request reviews** using **AI agents** in a sandboxed environment.
- **Designed asynchronous processing** with **Celery + Redis** for webhook-driven review workflows and issue-to-PR coding tasks.

Achievements

1st place at **Orange x Hexabot Conversational AI Hackathon** *Devpost*

- Built extensions for Hexabot visual editor, enabling multimodal capabilities like text-to-speech, speech-to-text, and vision.

2nd place at **IEEE TSYP 12 CS Technical Challenge** *Smartshield-INSAT*

- Earned 2nd place at the Tunisian Students and Young Professionals Computer Science Challenge by creating an AI-powered platform for automated cybersecurity threat detection and response, Smartshield.

Volunteering

Technical Manager *AINS 3.0*

Artificial Intelligence National Summit 3.0 (AINS) *Dec 2024 – Jun 2025*

- Led AINS 3.0 for IEEE Computer Society INSAT SBC: designed a three-track hackathon, organized a data science competition, coordinated speaker talks and panels, and co-organized four hands-on workshops spanning NLP and computer vision.

Skills

Languages: Python, TypeScript, Go, SQL

Tools & Technologies: Git, Linux, Docker, Kubernetes, NGINX, AWS, GCP, PostgreSQL, MongoDB, Redis, Qdrant, React, Next.js, FastAPI, Express.js, Hono, Bun, Celery, PyTorch, AI SDK, MCP, Langfuse